

Monthly Intelligence Report

Israel Edition 01 (Inaugural) — Tel Aviv & the Silicon Wadi

Where Israel operates an effectively insular grid with zero synchronous interconnects, ~70% gas + ~20% coal + ~10% renewable generation (Tamar + Leviathan offshore gas baseload, Orot Rabin + Rutenberg coal phase-out, solar-dominant Negev renewables), and the Dead Sea Transform Fault has been in a recognised "overdue major event" hypothesis cycle since the 1837 Galilee earthquake — the cohort-FIRST country with a transform-fault primary seismic source and a NEW R6_drought α anchor coupled to the desalination electricity-load multiplier. Inaugural edition · July 2026 · SSI v4.0.2

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SECTION A

Executive Summary

SUBSTATIONS SCORED

257

IEC TSO + private IPP
switchyards · 6 mehozot

GRID LINES MAPPED

~1,100

400 kV (limited) / 161 kV main /
22 kV / 12.6 kV

MC SIMULATIONS

2.57 M

10,000 per substation

PUBLIC DATA SOURCES

30+

95 variables · zero proprietary

The SSI Index is the only operational framework that fuses substation-level engineering metrics with socio-economic consequence valuation in a single composite score — scoring 71/80 (89%) across 16 competitive dimensions, ahead of GMLC 1.1 (48/80), EPRI (40/80), and academic MCDA frameworks (41/80). For Israel, its engine processes ~28,000+ source data points per run, executes 2.57 million Monte Carlo simulations through a 20×20 Gaussian copula correlation matrix, and performs ~310 million arithmetic operations — producing ~78,000+ output data points with full uncertainty quantification across 257 substations and ~1,100 grid lines spanning all 6 mehozot.

Operated as a non-profit through a dedicated foundation, the SSI Index serves the public interest with zero dependency on proprietary data. All 95 variables are drawn from 30+ verified public sources — IEC annual reports + transmission plan

2023–2030, Noga ILITO monthly + SMP wholesale market data (open since the November 2021 unbundling), PUA tariff resolutions + 2023–24 quality-of-service report, CBS regional accounts at the mehoz level, Bank of Israel macro + ILS FX, IMS climate + drought indices, GSI seismology + 475-yr PGA hazard maps along the Dead Sea Transform, INCD + CERT-IL critical-infrastructure scope, Mekorot national water-utility data + desalination-plant electricity-demand feeds (Sorek, Hadera, Ashkelon, Palmachim, Soreq B), MoEP air-quality monitoring — making the methodology fully auditable and reproducible. Israel is the inaugural single-country onboarding of Session 34 (post-CR), the first Middle-Eastern OECD member in the SSI v4 cohort, the first effectively-insular Mediterranean grid in cohort, and the first onboarding to introduce the NEW R6_drought α anchor and the precedent-setting R6_volcanic `active: false` flag pattern.

Substation in Focus

XX_100187

southern, southern · 161 kV

0.287

Medium

0.046

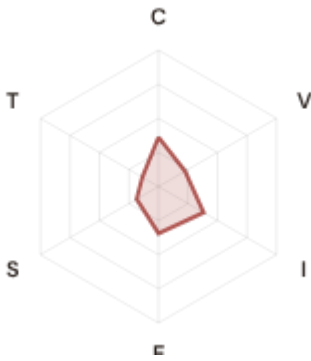
3th

R_FINAL

CLASSIFICATION

CI WIDTH

FLEET PERCENTILE



COMPONENTS

C

Continuity: 0.362 / 0.30 (121%)

I

Infrastructure: 0.383 / 0.25 (153%)

S

Saturation: 0.185 / 0.20 (92%)

V

Voltage: 0.225 / 0.10 (225%)

E

Economic: 0.344 / 0.10 (344%)

T

Transition: 0.137 / 0.05 (274%)

MODIFIERS

R3

Consequence: 0.921 ↓ dampens

R4

Graph Criticality: 0.955 ↓ dampens

R6a

Restoration: 1.070 ↑ amplifies

R6b

Seismic: 1.014 ↑ amplifies

R7

Digital Readiness: 0.985 ↓ dampens

This month's spotlight — automatically selected for June 2026 — is **XX_100187** in southern, southern. With an R_final of 0.287 (Medium band, 3th fleet percentile), its risk profile is dominated by the E (Economic) component at 344% of its weight ceiling, followed by T (Transition) at 274%. Modifiers collectively shift R_base by -20.0%, reflecting the substation's specific geographic, socio-economic, and network context.

SECTION B — RECURRING MODULE

Cyber & Economic Exposure Monitor

Why this section exists: Traditional grid risk frameworks treat cybersecurity and economic vulnerability as separate domains. The SSI's R7 (cyber posture) modifier and E (Economic) component allow us to cross these silos — revealing substations where digital-resilience headwinds intersect with high economic consequence. For Israel, the framing inverts the usual cohort narrative: the cyber posture is cohort-LEADING (R7 ceiling 1.040 via INCD 2017 + Unit 8200 IDF lineage since 1952 + CERT-IL 2014 + the Beersheba CyberSpark ecosystem with ~350+ private cyber companies and ~10% of the global cyber industry), and the relevant question is which mehozot combine that structural cyber strength with high consequence-of-loss in a continuous-active-threat environment — Hezbollah rocket campaign Nov 2023 – Nov 2024 ceasefire, Iran missile + drone barrages 13/14 April 2024 + 1 October 2024 (Arrow + David's Sling + Iron Dome interception), and the broader post-Oct-7 grid-as-target context absorbed via the `is\_high\_threat\_grid: true` country-config flag.

CYBER HIGH EXPOSURE

0

0.0% of fleet

BLIND SPOTS

1

High/Critical + R7 < median

AVG R7 MODIFIER

0.9805

Range [0.99, 1.040]

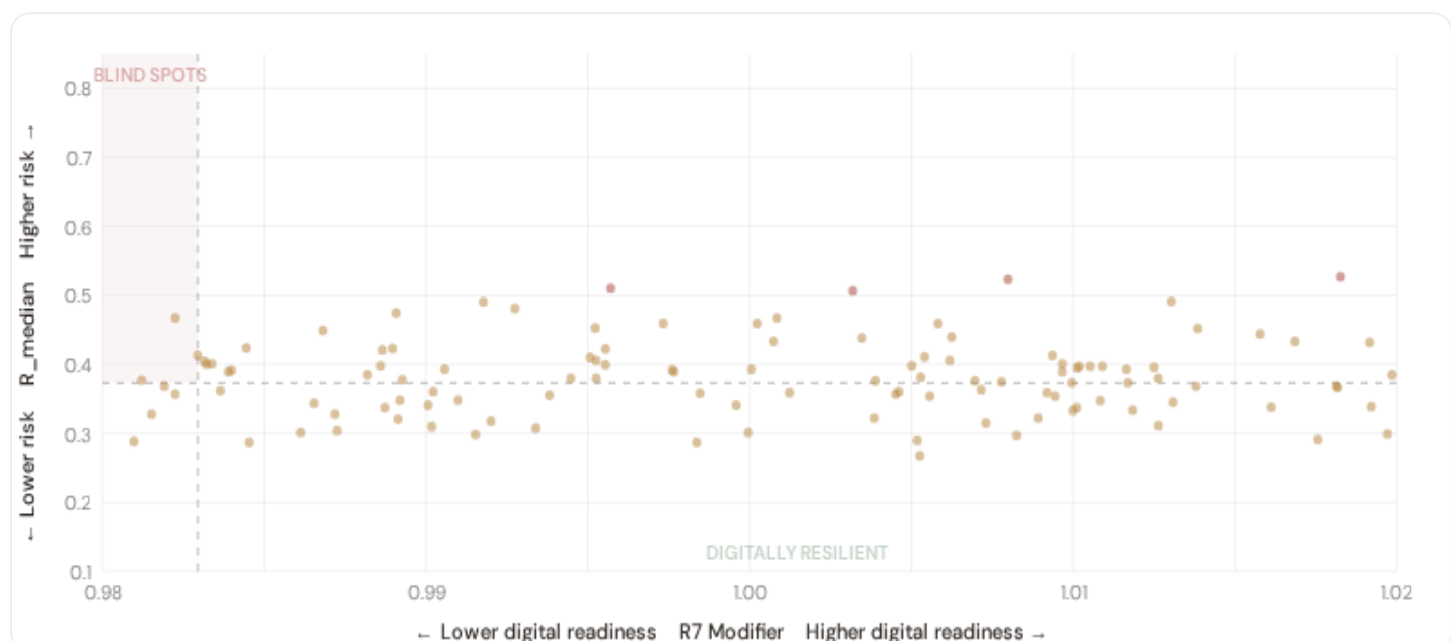
HIGH-CONSEQUENCE SUBS

65

25.3% · R3 ≥ 1.04

B.1 Cyber-Physical Risk Matrix

Each substation plotted by R7 cyber posture (x-axis) vs overall R_{median} (y-axis). The upper-left quadrant — high risk, low cyber posture — identifies *blind spots* where grid stress is high but monitoring/response capacity is weakest. For Israel the blind-spot population is structurally small: INCD coverage of designated critical-infrastructure entities (IEC, Noga, Mekorot, Bezeq, banks, defense contractors) is essentially nationwide, and the Beersheba CyberSpark + CERT-IL stack provides civilian-sector incident response across all 6 mehozot.



B.2 Economic Impact by Business Fabric

Low O

Med 62

High 3

Crit 0

Industrial-Tech-Logistics

Low O

Med 64

High O

Crit 0

 Government-Services

Low O

Med 61

High 3

Crit 0

 Periphery-Arid-Agriculture

Low O

Med 63

High 1

Crit 0

Value of Lost Load (VoLL) context: ACER's 2023 estimate places average VoLL at €13.1/kWh for industrial customers and €3.2/kWh for residential. The SSI's R3 consequence multiplier allows us to identify which substations serve the most economically exposed territories. **3 substations** combine capital-intensive economic fabric ($R3 \geq 1.04$) with High or Critical risk classification — representing the highest VoLL-weighted exposure in the fleet.

The capital-intensive tier concentrates in **southern (25), central (13), northern (13)** — regions where industrial corridors depend on uninterrupted power for continuous processes. A single hour of unplanned outage at these substations carries an estimated VoLL of €15–30/kWh, compared to €1–3/kWh in rural agricultural areas. The fleet-wide average energy poverty rate is 10.0%, but this masks sharp regional variation: Southern regions combine higher energy poverty with higher grid risk, creating a double vulnerability that the E component captures. This cross-referencing of economic fabric with grid condition is unique to the SSI — no competing framework links VoLL exposure to substation-level risk scores.

B.3 Mehoz Cyber Posture Ranking

The 6 mehozot ranked by average R7 modifier — for Israel, lower values are rare given the nationwide INCD coverage and the Unit 8200 talent-pipeline footprint. The differentiator is sector concentration: Tel Aviv + Central host the Silicon Wadi cyber HQ cluster (Check Point + CyberArk + Wiz + Armis + dozens of mid-cap cyber-pure-plays), while Southern (Beersheba CyberSpark) hosts CERT-IL and the Ben-Gurion University cyber-engineering programme. Cross-referenced with average R_{median}, the Negev periphery carries the largest grid-stress headwind via the R6_{drought} desalination multiplier and the longer 22 kV radial feeds to Mitzpe Ramon and outlying Bedouin settlements.



WORST DIGITAL READINESS — TOP 15 MEHOZ REGIONS

Longer bar = higher R7 = better digital infrastructure.

1	central		0.9751
2	northern ▲ high R		0.9793
3	jerusalem ▲ high R		0.9801
4	southern		0.9809
5	haifa ▲ high R		0.9812
6	tel-aviv		0.9966

Mehoz-level analysis reveals a clear digital divide mirroring the broader DESI pattern. **central** has the lowest average R7 (0.9751), while **tel-aviv** leads at 0.9966 — a spread of 0.0215 across the R7 range. **3 mehoz regions** combine below-median digital readiness with above-median grid risk, making them priority targets for digitalisation investment. The R7 modifier is intentionally narrow to reflect the current limitations of open DESI data — as NIS2 compliance reporting matures, future editions will expand R7's dynamic range and incorporate operational technology (OT) security indicators.

B.4 Monthly Pulse

Inaugural edition — Baseline Snapshot (June 2026). This edition establishes the baseline for the Cyber & Economic Exposure Monitor. Key reference points: fleet average R7 = 0.9805, median = 0.9830; 0 substations classified HIGH cyber-exposure; 1 blind-spot substations (High/Critical risk + below-median R7); 0 Critical-band substations with below-median digital readiness. Future editions will track deltas against this baseline.

SECTION C

Data Refresh & Changelog

The SSI v4.0.2 draws on **13 verified public data sources** feeding 95 variables across 6 components and 7 modifiers. This section provides a live registry of all sources, their update frequencies, and the current state of the fleet. Transparency in data vintage is central to the SSI's credibility.

TOTAL SOURCES

13

95 variables

WEEKLY / MONTHLY

3

High-frequency feeds

QUARTERLY / ANNUAL

7

Regular refresh cycle

STATIC / DERIVED

0

Standards + scenarios

Data Source Registry — June 2026

d01_iec	IEC — Israel Electric Corporation (TSO + dominant generator + dominant DSO; state-owned ~99.8%; founded 1923 as Palestine Electric Company, rebranded 1961)	DAILY	C2,C3,I1,I2,I3,R4	14 vars
d03_gsi	GSI — Geological Survey of Israel (Seismology Division operates national seismic network)	DAILY	I1,I2,R6a_seismic	10 vars
d06_ims	IMS — Israel Meteorological Service (under Ministry of Transport; runs ECMWF mirror + national observation network)	DAILY	I1,I2,I3,R6_climate,R...	8 vars
d01b_noga	Noga ILITO — Israel Independent System Operator (carved out from IEC 1 Nov 2021 under June 2018 reform)	CONTINUOUS	I1,I3,S1,T1	8 vars
d05_osm	OSM Overpass — grid topology (ISO3166-1=IL area filter; effectively insular grid)	MONTHLY	I1,I2,I3,R4	8 vars
d07_copernicus	Copernicus ERA5 + CMIP6 (reanalysis + climate projections)	MONTHLY	R6_climate,R6_dro...	4 vars
d09_incd_certil	INCD — Israel National Cyber Directorate (PMO, 2017) + CERT-IL (2014, CyberSpark Beersheba) + Unit 8200 (IDF, 1952)	CONTINUOUS	R7_cyber	8 vars
d12_mekorot	Mekorot — national water utility + desalination plant electricity-demand data (NEW source for R6_drought)	MONTHLY	R6_drought,E1	6 vars
d02b_dso	Kibbutz / moshav cooperative DSO aggregate (~10% of distribution; rural electrification co-ops carved out historically)	QUARTERLY	C1,V1,I2	4 vars
d04_cbs	CBS — Central Bureau of Statistics (Halamas; national statistical office, demographics + national accounts + mehoz)	QUARTERLY	E1,E2,S1,S2,S3	16 vars
d04b_boi	Bol — Bank of Israel (national accounts, FX ILS, macro projections)	QUARTERLY	E1,E2	4 vars
d02_pua	PUA — Public Utilities Authority (Electricity), since 1996 (independent statutory regulator; MEDREG member)	ANNUAL	C1,C2,V1,V2,E1	12 vars
d10_iaeoecd	IEA + OECD — energy benchmarks (Israel 33rd OECD member, accession 7 Sep 2010)	ANNUAL	T1,E1	6 vars

C.1 Update Frequency Timeline

EXPECTED UPDATE WINDOWS

Monthly (3 sources)

Grid load + procurement data

OSM Overpass · Copernicus ERA5/CMIP6 · Mekorot (desalination demand)

Quarterly (3 sources)

DER registry + business + macro

CBS (Halamas) · Bank of Israel · Kibbutz/moshav cooperative DSO aggregate

Annual (4 sources)

Core grid quality + demographics

PUA (Public Utilities Authority) · IEA/OECD · MEI (Ministry of Energy and Infrastructure) · OECD Going Digital

C.2 Fleet Score Snapshot

FLEET SIZE

257

substations scored

MEDIAN R

0.373

P5=0.288 · P95=0.483

HIGH + CRITICAL

7

2.7% of fleet

HIGH CONFIDENCE

99%

255 subs · CI/R < 0.50

BAND DISTRIBUTION

Low 0 (0.0%)

Medium 250 (97.3%)

High 7 (2.7%)

Critical 0 (0.0%)

No primary data sources have been updated since the v4.0.2 launch baseline. All **257 substation scores remain unchanged** this edition. The fleet median R of 0.373 (mean 0.375) reflects a distribution skewed toward the lower bands, with 0.0% in Low and 97.3% in Medium. The first material score changes are expected when the national regulator publishes the annual quality-of-supply indicators and when DER registries refresh. High-frequency sources (OSM, weather) update weekly but feed only infrastructure topology (R4) and thermal proxy (I5), which have minimal impact on fleet-level score distribution.

C.3 Changelog — Israel Edition 01 (Session 34)

9 changes tracked across PO, PI, and P2 releases

IL-S34-9	ENHANCED	Greenfield thin-shell on post-S33 D#21-codified architecture — 5 core pages (index/regional/map/data + ssi-metadata.js) authored directly on CountryRenderer.Safe + normalizeMeta() (KB §68.10/11) + admin-unit-suffix tolerance (KB §69.11) + content-leakage gate (KB §73.x BPG Discipline #21)	KB §74
IL-S34-8	NEW	Section D deepDives rotation — 6 Israeli mehozot rotation seeded with Tel Aviv (Edition 01 inaugural anchor — Silicon Wadi) + Haifa (Bazan refinery + Carmel chemical complex + Yokne’am chip cluster Mellanox/Nvidia) + Central (Sharon Plain tech + Sorek/Palmachim desalination cluster) + Southern (Negev + Beersheba CyberSpark + Dimona + R6_drought peak) + Jerusalem (capital + Hebrew University + Mobileye HQ + DST seismic) + Northern (Galilee agriculture + Migdal HaEmek)	intelligence D
IL-S34-7	ENHANCED	Edition patcher anchored at FIRST_REFRESH 2026-08-13 — single-country onboarding (Session 34) post-S33 D#21 codification; edition_anchor_month_offset = 5 per cohort sync with SI/SK/HU/IS/KR/CR (Edition 02 = 2026-07-09)	intelligence
IL-S34-6	NEW	ESG report — israel entry in COUNTRY_SOURCES (13 IL-specific references including IEC + PUA + Noga ILITO + GSI 1927 Jericho + CBS end-2024 + Mekorot desalination + INCD + CERT-IL)	esg-report
IL-S34-5	NEW	R3 4-tier calibration — Tel Aviv 1.06 / Central + Haifa 1.05 / Jerusalem 1.04 / Northern + Southern 1.03 — moderate-upper gradient cohort mid-upper (GDP/cap spread Tel Aviv \$75-90k district-level to Northern/Southern \$35-45k district-level)	methodology
IL-S34-4	NEW	R6_drought NEW α anchor — FIRST in cohort — arid-climate drought + desalination-electricity nexus modifier (Negev 0.16-0.18 + Central coastal-desalination 0.12-0.14 + Northern Galilee 0.08-0.10); R6_volcanic INACTIVE flag (Pliocene Golan basalts ~4 Mya only); R6a_seismic Dead Sea Transform strike-slip primary source (1927 Jericho Mw 6.2 + 1995 Aqaba Mw 7.2 anchors)	methodology
IL-S34-3	DATA	d05_osm LIVE — 257 substations via ISO3166-1=IL area filter (effectively insular grid; zero synchronous interconnect; EuroAsia HVDC IL-CY-GR Phase 1 in construction target 2027-2028 documented forward-only)	methodology
IL-S34-2	DATA	IEC (dominant TSO + dominant generator + dominant DSO ~80%) + Noga ILITO (independent system operator since 1 Nov 2021) + PUA regulator (since 1996) + MEI policy + CBS statistics + Bank of Israel macro + GSI seismic + IMS climate + Mekorot desalination + INCD + CERT-IL + Unit 8200 wired	methodology
IL-S34-1	ENHANCED	KB v34 §74 — Israel inaugural onboarding (Session 34, single-country post-S33 D#21 codification; ILS currency post-symbol; first Middle-Eastern OECD member; OECD accession 7 Sep 2010 as 33rd member; first effectively-insular Mediterranean grid; first R6_drought modifier; first R6_volcanic INACTIVE precedent; R7 ceiling 1.040 cohort-LEADING)	KB §74

SECTION D — MONTHLY DEEP-DIVE

Tel Aviv — Silicon Wadi High-Tech Corridor & the R3 1.06 Anchor

Deep-dive rotation: Each edition features a substantive thematic analysis. The 6-mehoz rotation covers Israel's most material narratives: **Ed. 01** Tel Aviv — Silicon Wadi HQ corridor (Check Point, Wix, Monday.com, Fiverr, Mobileye listings, TASE) · the R3 1.06 capital-tech anchor · **02** Haifa — Bazan oil refinery + Carmel chemical complex + Yokne'am chip cluster (Mellanox/Nvidia Networking, Marvell) + Technion · **03** Central — Sharon Plain tech (Herzliya, Ra'anana, Petah Tikva — CyberArk HQ) + Rosh HaAyin + Sorek/Palmachim desalination cluster · **04** Southern — Negev arid corridor + Beersheba CyberSpark + Dimona Nuclear Research + Dead Sea Works (potash, bromine, magnesium) + Eilat + R6_drought peak · **05** Jerusalem — government services + Knesset + Bank of Israel + Hebrew University + Mobileye HQ + Hadassah + Dead Sea Transform northern segment seismic anchor · **06** Northern — Galilee agriculture + Migdal HaEmek tech satellite + Kiryat Shmona + Druze + Arab Israeli municipalities + Hula Valley DST exposure.

D.1 Why Tel Aviv, Why Now

CORRIDOR SUBSTATIONS

18

18 EHV · 0 HV

HIGH BAND

0

0.0% of corridor

CORRIDOR MEDIAN R

0.373

Fleet median: 0.373

WORST MEHOZ

tel-aviv

Avg R: 0.366 · 18 substations

The Tel Aviv — Silicon Wadi high-tech corridor (Edition 01 inaugural anchor) hosts **18 substations** across 1 mehoz regions, making it the locus of this edition's deep-dive analysis. Its risk profile is concentrated in the upper bands: **0 substations (0.0%)** are classified High, with only **0** in the Low band. 39% of corridor substations score above the national fleet median of 0.373. The corridor median R of 0.373 reflects the local risk profile. The corridor concentrates the country's industrial heartland. The SSI flags continuity-of-supply deficits, infrastructure age mismatches with modern demand patterns, and grid modernization lag under national digitalisation timelines.

D.2 Corridor Map and Scoring



SUBSTATION	MEHOZ	R_FINAL	BAND	C	V	I	E	S	T	ALERT
תחנת משנה ירקון	tel-aviv	0.418	Medium	0.527	0.405	0.407	0.410	0.189	0.122	
XX_100015	tel-aviv	0.415	Medium	0.527	0.474	0.283	0.418	0.201	0.119	
XX_100061	tel-aviv	0.410	Medium	0.361	0.466	0.391	0.433	0.202	0.085	
XX_100030	tel-aviv	0.406	Medium	0.515	0.405	0.337	0.289	0.202	0.119	
חטיבת ייצור והולכה - תחנת רמת השרון	tel-aviv	0.395	Medium	0.411	0.451	0.272	0.366	0.186	0.128	
XX_100010	tel-aviv	0.386	Medium	0.463	0.432	0.448	0.266	0.165	0.091	
תחמ"ג איילון	tel-aviv	0.375	Medium	0.322	0.422	0.439	0.318	0.219	0.081	
תה"ר הולץ	tel-aviv	0.373	Medium	0.507	0.251	0.254	0.357	0.197	0.078	
XX_100121	tel-aviv	0.373	Medium	0.455	0.334	0.410	0.394	0.227	0.077	
תחנת משנה רמת-גן	tel-aviv	0.357	Medium	0.339	0.300	0.445	0.384	0.212	0.077	

... 8 additional substations — [explore full corridor in Digital Twin →](#)

D.3 Component Analysis

COMPONENT AVERAGES — TEL AVIV — SILICON WADI HIGH-TECH CORRIDOR (EDITION 01 INAUGURAL ANCHOR) VS FLEET			MODIFIER AVERAGES — TEL AVIV — SILICON WADI HIGH-TECH CORRIDOR (EDITION 01 INAUGURAL ANCHOR) VS FLEET		
C — Continuity	0.433 vs 0.441 (−2%)		R3 Consequence	1.009	fleet 1.001 →
I — Infrastructure	0.339 vs 0.350 (−3%)		R4 Graph Crit.	0.970	fleet 0.981 ↓
S — Saturation	0.195 vs 0.233 (−16%)		R6a Restoration	1.036	fleet 1.025 ↑
V — Voltage	0.370 vs 0.361 (+3%)		R6b Seismic	1.097	fleet 1.109 ↑
E — Economic	0.354 vs 0.350 (+1%)		R7 Digital Read.	0.997	fleet 0.981 →
T — Transition	0.097 vs 0.107 (−10%)				

The dominant risk driver across the Tel Aviv — Silicon Wadi high-tech corridor (Edition 01 inaugural anchor) corridor is **V (Voltage)**, averaging 0.370 against a fleet average of 0.361 — occupying 370% of its weight ceiling (w = 0.10). The second contributor is **E (Economic)** at 0.354 vs fleet 0.350 (354% of ceiling). Among modifiers, the R3 consequence multiplier averages 1.009 (fleet: 1.001), reflecting the socio-economic exposure of the corridor. The R6b seismic modifier averages 1.097, capturing the local seismic exposure — a dimension invisible to every competing framework.

D.4 Mehoz Breakdown

MEHOZ RISK PROFILE — SORTED BY MEAN R

Longer bar = lower R = better resilience.

tel-aviv

18 subs · avg R = 0.366 · H:0 M:18

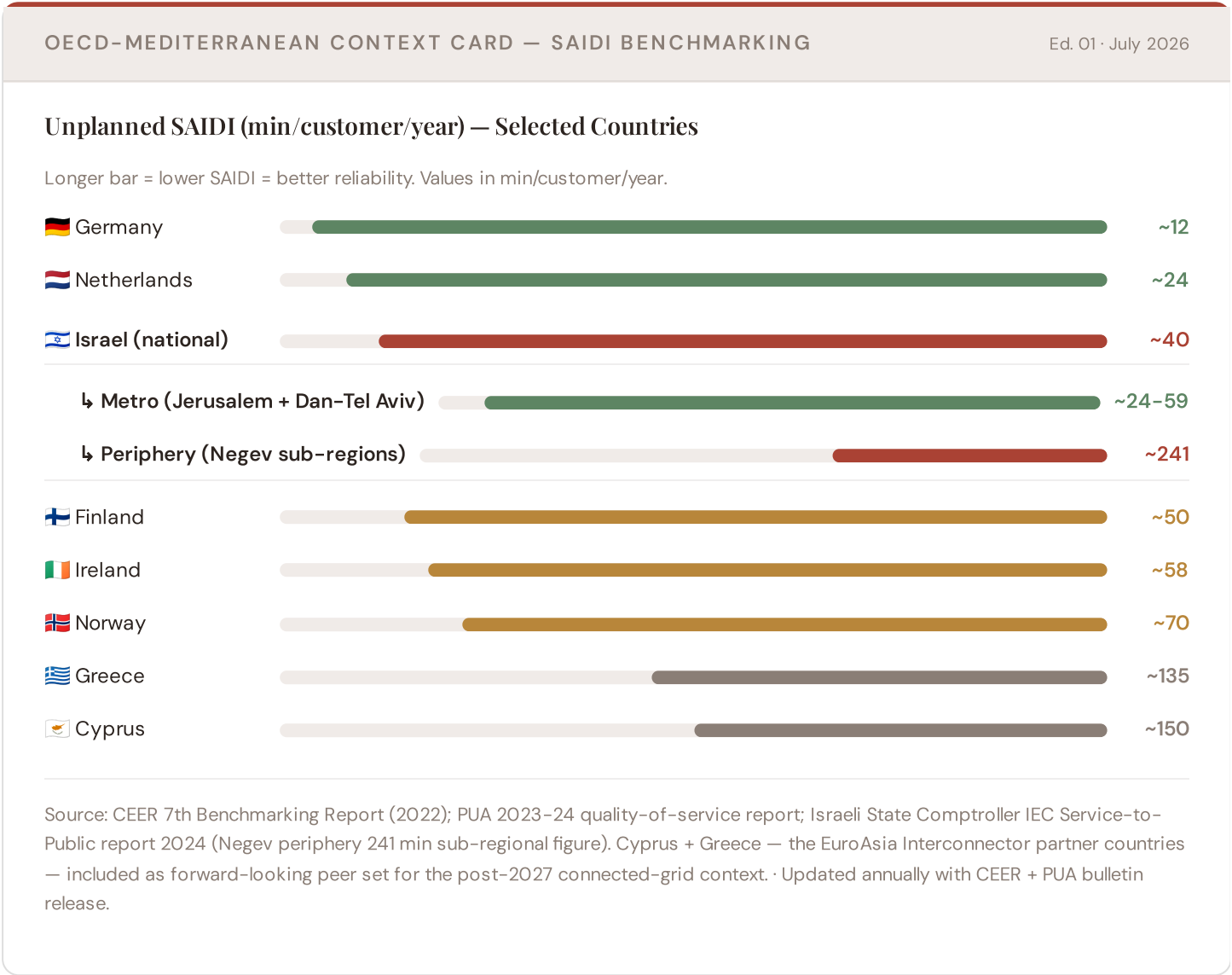
The mehoz regions-level breakdown reveals significant intra-regional variation. **tel-aviv** leads with a mean R of 0.366 across 18 substations, while **tel-aviv** scores 0.366 — a spread of 0.000 within a single corridor. The SSI's substation-level granularity reveals that the Tel Aviv — Silicon Wadi high-tech corridor (Edition 01 inaugural anchor) is not uniformly stressed but contains distinct sub-corridors of concentrated risk — precisely the kind of spatial pattern that investment planning requires.

D.5 Implications

O Tel Aviv — Silicon Wadi high-tech corridor (Edition 01 inaugural anchor) substations score $R \geq 0.650$, placing them in the upper quartile of national risk. For NEPN / national digital plan / RRP investment prioritisation, this analysis identifies the tel-aviv sub-corridor as the highest-priority target — where DER deployment is accelerating against ageing infrastructure with above-average continuity deficits and significant socio-economic exposure. The SSI's silo-breaking approach reveals what no single-discipline framework can: that these substations matter not only because of their engineering condition, but because the communities they serve are disproportionately vulnerable to disruption. This is precisely the anticipatory grid planning capability that the SSI was built to provide.

OECD-Mediterranean Context Card

How this works: Each edition includes one peer-benchmarking card. The card is selected to complement the month's deep-dive theme. This month: **SAIDI comparison** — because the Tel Aviv-corridor analysis hinges on Israel's continuity performance relative to insular and Mediterranean peers, especially Cyprus + Greece (the EuroAsia partner countries) and Ireland + Finland (insular / peninsular cohort siblings). Future editions rotate through: VoLL comparison (Ed. 02), DER saturation vs Mediterranean peers (03), CMIP6 drought hazard vs arid peers (04), etc. The underlying data updates annually with the CEER + PUA + State Comptroller reliability bulletin cycle.

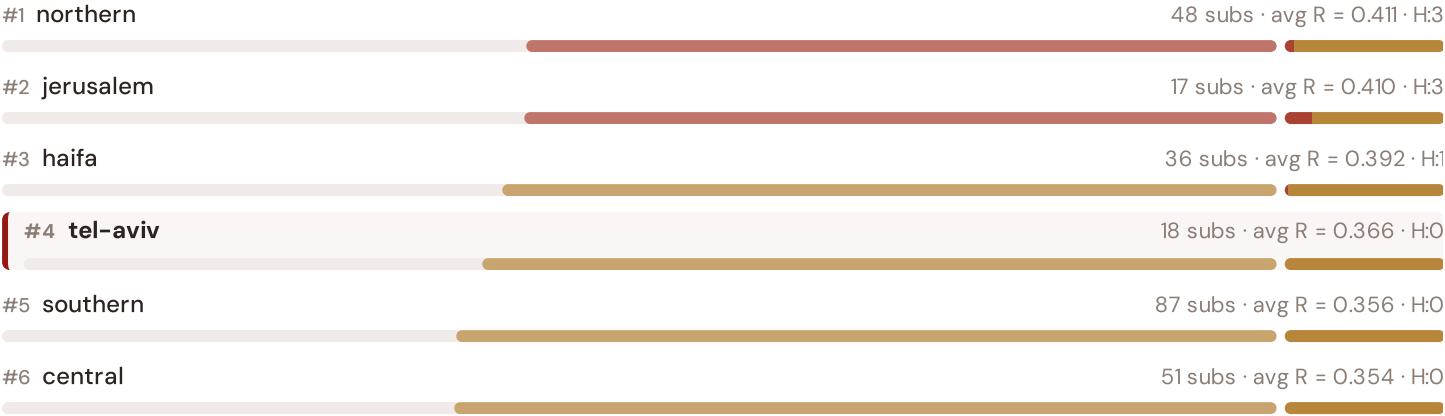


This month's key insight: The Tel Aviv — Silicon Wadi high-tech corridor (Edition 01 inaugural anchor) corridor deep-dive shows **18 substations** (of 18) with C-component above 70% of its weight ceiling — indicating continuity-of-supply performance consistent with rural territory classification. The corridor's average C of 0.433 is **1.0× the fleet average** of 0.441. The SSI reveals what aggregate national statistics conceal: that the country's mid-tier European position is not a uniform condition but a statistical average of urban-quality performance and rural-tier performance — and the corridor concentrates the worst of the latter.

Regional Risk Ranking — All 6 Israeli mehozot

Where does Tel Aviv sit within the national landscape? Mehozot sorted by mean R_median, with band distribution and fleet comparison.

Longer bar = lower R = better resilience. Sorted worst → best.

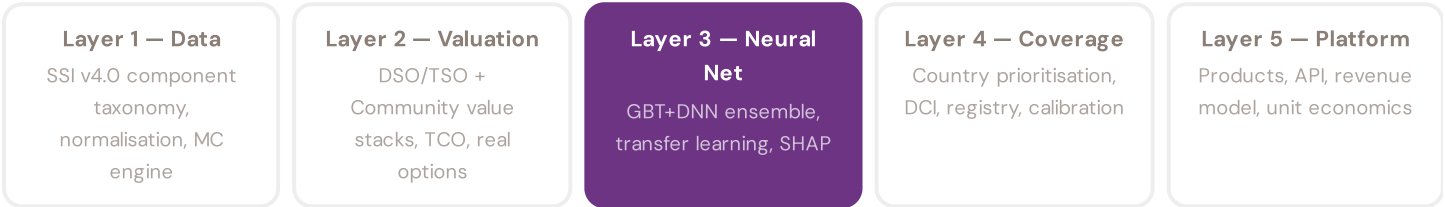


The Tel Aviv — Silicon Wadi high-tech corridor (Edition 01 inaugural anchor) ranks **#4 of 6 mehoz regions** by mean R_median, placing it among the upper tier of national risk. The three most stressed are northern, jerusalem, haifa, while the three most resilient are central, southern, tel-aviv. 3 of 6 score above the fleet average of 0.375. This ranking — possible only because the SSI scores every substation individually rather than relying on aggregate DSO or national statistics — reveals the true spatial distribution of grid stress. It is precisely this substation-level granularity that positions the SSI as a tool for investment prioritisation: not treating entire regions as monoliths, but identifying the specific corridors within each that demand attention.

SECTION F — RECURRING MODULE

SSI Enhanced Neural Network — Monthly Topic

From grid intelligence to BESS valuation platform. The SSI Index answers *"How healthy is this substation?"* The SSI Enhanced Neural Network (SSI-ENN) extends this into: *"What is a BESS worth at this substation — to IEC as the dominant utility, to Noga ILITO as the system operator, to the IPP cluster (Dorad, OPC, Edeltech, Bazan, Dalia), to PUA as the tariff regulator, and to the community?"* Each edition explores one building block of the SSI-ENN's five-layer architecture, showing how the SSI Index data feeds directly into investment-grade grid analytics — particularly relevant for Israel given the 30%-renewable-by-2030 target (~180 B / ~\$22 B plan) requires significant storage to integrate the planned ~15 GW of new solar.



LAYER 3 §0.5.6 · June 2026

Explainability & SHAP Values

Clients need to understand not just the prediction but WHY a substation ranks highly

Investment-grade models require interpretability. SHAP (SHapley Additive exPlanations) provides a rigorous, game-theoretic attribution of each feature's contribution to each prediction. For every substation, the SSI-ENN produces a SHAP waterfall: starting from the fleet-average valuation and showing how each feature pushes the estimate up or down. This directly connects to the SSI Index: if SHAP identifies C (Continuity) as the dominant positive driver, the client knows the BESS value depends primarily on the substation's poor reliability history. Transparency converts model outputs into investment decisions.

KEY CONCEPTS

→ SHAP: game-theoretic feature attribution per prediction

→ Waterfall: fleet average → substation prediction

→ Direct traceability to SSI components and data sources

→ Feature importance validated against domain expertise

LIVE SSI DATA CONNECTION

The training set comprises 257 substations with complete feature vectors. Average CI_width of 0.083 across the fleet provides the uncertainty ground truth that the neural network's quantile regression heads must calibrate against. The fleet's risk distribution — 0 Low, 250 Medium, 7 High, 0 Critical — ensures balanced training across all risk bands.

Coming next month:

LAYER 3

 Uncertainty Quantification — *Investment-grade predictions require calibrated confidence intervals, not just point estimates*

SECTION G

Looking Ahead

NEXT MONTH — EDITION 02

Deep-Dive: southern

Deep-dive into southern's grid risk profile — substation map, component analysis, mehoz regions breakdown.

NEXT DATA REFRESH

Q3 2026 — 3 Quarterly Sources

CBS (Halamas), Bank of Israel, Kibbutz/moshav cooperative DSO aggregate are due for refresh in Q3. Impact: DER registry (T1), business fabric indicators (E2), macro-economic data. Highest-impact annual source pending: **PUA — Public Utilities Authority (Electricity), since 1996 (independent statutory regulator; MEDREG member)** (12 variables → C1,C2,V1,V2,E1).

FLEET SPOTLIGHT

jerusalem: Highest Risk Concentration

jerusalem has the highest concentration of High/Critical substations: **3 of 17 (18%)** are in the upper risk bands — avg $R = 0.410$. This makes it the most acute risk corridor in the fleet.

Edition 02 will be published on the second Thursday of July 2026. Deep-dive region: **southern**. To ensure you receive it, confirm your subscription segment (General / Institutional / Academic) by email to ssi_index@ikenga.eu.

Feedback welcome. The SSI Index is designed to evolve through stakeholder dialogue. If you represent IEC, Noga ILITO, PUA, the Ministry of Energy and Infrastructure, INCD, CERT-IL, CBS, Bank of Israel, GSI, IMS, Mekorot, Dorad, OPC Energy, Edeltech, Bazan, Dalia, the Technion, the Hebrew University, the Weizmann Institute, Ben-Gurion University CyberSpark, or any Israeli academic or industry institution and would like to contribute data or methodology feedback, please contact us at ssi_index@ikenga.eu. The SSI's credibility depends on open scrutiny.

SSI Monthly Intelligence Report — Israel Edition 01 (Inaugural)

Published 9 July 2026 · SSI Index v4.0.2 · 257 substations · ~1,100 grid lines · 6 components · 20 metrics · 7 modifiers (incl. R6_drought NEW + R6_volcanic INACTIVE)

10,000-iteration Gaussian Copula Monte Carlo · Sobol global sensitivity analysis (196,608 evals/substation) · CMIP6 SSP2-4.5